# Joshua Xu

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## Education

**University of British Columbia** Graduated May 2026

*BASc Electrical Engineering, First Class Honours, Dean's List 2023–2024 | GPA: 3.85/4.33* *Vancouver, BC*

- **Relevant Courses**: Power Electronics (82%), Analog CMOS IC Design (92%), Micro/Nanofabrication Lab (80%), Antennas and Propagation (83%)

## Technical Skills

**Languages & Scripting**: Python, C, 8051 Assembly, SystemVerilog
**Tools & Lab**: Altium Designer, LTspice, Cadence Virtuoso/Spectre, Minitab, Oscilloscope, DC Power Analyzer
**Quality & Methods**: SPC, Cpk, MSA/Gage R&R, RCA, DFM

## Experience

**Quality Assurance Technician** May 2026 – Present

*International Water-Guard Industries Inc. – aerospace manufacturing, ISO 9001 / AS9100, MRB* *Surrey, BC*

- Supported closure of 50+ nonconformances in inherited **AS9102** FAI reports and inspection records, re-measuring parts to restore missing data; Airbus accepted both products at source inspection
- Spearheaded **AI** cross-checks of 100+ FAIs against drawings and CoCs for Airbus, cutting a week of review to hours
- Write the technical basis for **MRB** dispositions on about 5 NCRs a week; on two obsolete pumps in a $7.5k assembly, argued for rework over replacement, found Teflon debris left from assembly, and the pumps then worked
- **Trained** a new team member on inspection and NCR procedures over 3 months; now fully independent

**Advanced Quality Engineer Intern** Sep. 2024 – Aug. 2025

*EnerSys Inc. (Alpha Technologies Ltd.) – telecom DC power, SPC, Cpk, DFM, FAI* *Burnaby, BC*

- Qualified the production test station for a 5 kW rectifier module across 30 parameters; traced 9 failures to a wrongly set limit (one parameter **Cpk** 0.52 to 1.65); my report released it to the contract manufacturer
- Built a **Pareto** of 47 field returns of a 4.6 kW rectifier: MOSFET failures led (13 of 47), ahead of PFC-stage and capacitor faults
- Led a failure analysis **teardown** of a third-party module, finding commercial-grade parts in an outdoor product, and raised it to the director as a sourcing risk
- Caught insufficient **solder mask** width between vias and pads, a shorting risk, in DFM reviews of 30+ assemblies including PCBAs; engineers moved parts to meet the 0.1 mm minimum
- Ran 6+ **Minitab** studies (SPC, capability, Gage R&R) and wrote the MSA training module for technicians at the Burnaby and Reading, PA sites

## Projects

**Marine Energy Harvester: BQ25570 Power Stage** ↗ | *Altium, Python* Sep. 2025 – Apr. 2026

- Owned the power subsystem in a 5-person capstone (85%) supporting NATO SPS Project SHIELD; hosted the weekly meeting, and both **board revisions** finished on schedule
- Chose the **BQ25570** over the ADP5090 and BQ25504 for its 100 mV minimum input and integrated buck, accepting 488 nA quiescent current; laid out my section of the multilayer PCB
- On bring-up my stage left cold start at ∼1.8 V and held storage at ∼3.0 V; my **validation plan** showed system efficiency peaking at ∼34% vs ≥30% required
- Traced a two-week **regulation failure** to floating gates upstream; the owner's fix restored regulation

**Reflow Oven Controller** ↗ | *8051 Assembly, Python, SPI* Mar. 2023

- Wrote 8051 assembly **FSMs** with PWM relay drive and built the thermocouple front end (OP07, MCP3008 over SPI); the oven ran the full profile, with readings within about ±3 °C of a reference thermometer

**Magnetic-Field-Controlled Robot** ↗ | *C, EFM8, PIC32, I2C* Apr. 2023

- Wrote C firmware for an EFM8 transmitter and a **PIC32** receiver decoding frequency-coded commands; fixed MOSFET burn-outs and a noisy ADC, and it responded at up to 130 cm

## Additional Information

**Work Status**: Canadian Citizen; eligible for TN status (U.S.); open to relocation in Canada and the U.S.
**Certifications**: IPC-A-610
**Languages**: English, Conversational Mandarin
**Interests**: Badminton, piano, reading